

DIGITAL SIGNAL PROCESSING

COURSE PROJECT REPORT

Project I: ECG Signal Denoising for Telemedicine Applications
Project II: Multi-Band Speech Equalizer for Podcast Enhancement

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Abstract

This report presents the design and implementation of two digital signal processing applications developed as part of the DSP course project. **Project I** focuses on ECG signal denoising for telemedicine use, where multiple digital filters — including FIR windowed filters and IIR Butterworth and Chebyshev designs — are applied to MIT-BIH arrhythmia database recordings to suppress baseline wander, power-line interference, and EMG noise. All filters are evaluated through frequency response analysis, SNR improvement metrics, and spectral comparison. **Project II** implements a multi-band speech equalizer for podcast enhancement, featuring seven frequency bands with adjustable per-band gains, both FIR (Blackman window) and IIR filter options, and support for custom band configurations and variable output sample rates. Both projects demonstrate practical DSP design skills, from filter specification through implementation and quantitative validation.

PROJECT I

ECG Signal Denoising for Telemedicine Applications

I. Problem Formulation

A. ECG Signal Characteristics

The electrocardiogram (ECG) is a clinical measurement of the electrical activity of the heart, widely used in telemedicine and wearable health monitoring systems. For this project, ECG recordings are sourced from the MIT-BIH Arrhythmia Database hosted on PhysioNet, sampled at 360 Hz. Two records are used: Record 100 (normal sinus rhythm) and Record 106 (ventricular ectopic beats), providing both a clean reference and a clinically challenging signal.

The useful diagnostic ECG bandwidth, as defined by AHA (American Heart Association) guidelines, spans 0.5 to 100 Hz. This range captures all clinically significant waveform components including the P wave, QRS complex, and T wave, which carry information about atrial and ventricular activity, conduction timing, and cardiac health.

B. Noise Components

Three primary noise sources corrupt clinical ECG signals:

- Baseline wander (< 0.5 Hz): caused by respiration and patient movement, producing slow amplitude drift that shifts the ECG baseline and can obscure ST-segment changes.
- Power-line interference at 50 Hz: introduced by electrical equipment in clinical environments, appearing as a sinusoidal artifact superimposed on the signal.
- EMG (muscle) noise (20–150 Hz): generated by skeletal muscle activity, producing high-frequency noise that obscures fine waveform details, especially during patient movement.

C. Filtering Objectives

The filtering pipeline is designed with three sequential stages:

- High-pass filter (HPF) at 0.5 Hz to remove baseline wander while preserving the low-frequency P-wave content.
- Notch filter at 50 Hz to eliminate power-line interference with minimal effect on nearby ECG components.
- Low-pass filter (LPF) at 100 Hz to suppress high-frequency EMG noise while retaining the full diagnostic bandwidth.

II. Filter Design Specifications

A. Specification Parameters

The following specifications were selected based on clinical ECG standards and the MIT-BIH database characteristics:

- Passband ripple (Rp): 0.5 dB — limits amplitude distortion within the diagnostic band.
- Stopband attenuation (Rs): 40 dB — ensures adequate noise suppression across all filter stages.
- HPF cutoff frequency: 0.5 Hz — boundary between baseline wander and useful ECG content.
- LPF cutoff frequency: 100 Hz — upper limit of the diagnostic ECG band.
- Notch frequency: 50 Hz — target frequency for power-line interference removal.
- Notch Q factor: 25 — gives a 2 Hz bandwidth centered at 50 Hz, balancing attenuation depth and bandwidth impact.

B. Implementation Constraints

- Filter order: FIR order 1000 was chosen to achieve adequate stopband attenuation at the very low HPF cutoff ($0.5 \text{ Hz} / 360 \text{ Hz} = \text{normalized } 0.00278$). This is appropriate for offline analysis.
- Latency: FIR latency is approximately 1389 ms, which makes it unsuitable for real-time monitoring. IIR filters (order 4–20) introduce negligible latency and are preferred for real-time systems.
- Computational complexity: FIR requires 1001 multiplications/sample. IIR Butterworth LPF (order 20, SOS) requires approximately 41 multiplications/sample.
- Numerical stability: High-order IIR filters are implemented in Second-Order Sections (SOS) form using `tf2sos()` to prevent numerical overflow and coefficient quantization errors in direct-form implementations.
- Phase distortion: All filtering uses `filtfilt()` to achieve zero-phase (forward-backward) filtering, eliminating group delay distortion that would shift ECG waveform features in time.

III. Filter Design and Analysis

A. Summary of Designed Filters

Filter	Type	Order	Cutoff	Atten.	Purpose
FIR Kaiser HPF	FIR (Kaiser)	1000	0.5 Hz	~44 dB	Baseline wander
FIR Hamming HPF	FIR (Hamming)	1000	0.5 Hz	~41 dB	Baseline wander
Butterworth HPF	IIR Butterworth	4	0.5 Hz	60 dB	Baseline (smooth)
Chebyshev HPF	IIR Cheby I	4	0.5 Hz	69 dB	Baseline (sharp)
Notch Filter	IIR Notch Q=25	2	50 Hz	40.2 dB	Power-line 50 Hz

Filter	Type	Order	Cutoff	Atten.	Purpose
Butterworth LPF	IIR Butterworth	20	100 Hz	47.9 dB	EMG suppression
Chebyshev LPF	IIR Cheby I	15	100 Hz	86.1 dB	EMG suppression
FIR Kaiser LPF	FIR (Kaiser)	1000	100 Hz	96.8 dB	EMG suppression

B. FIR Filters (Window Method)

Kaiser Window (HPF and LPF, Order 1000)

The Kaiser window is applied with a shape parameter $\beta = 6$, providing approximately 44 dB of stopband attenuation and a flexible trade-off between transition bandwidth and ripple. The high-pass filter is designed with a normalized cutoff of $0.5/180 = 0.00278$, which represents an extremely narrow normalized transition band, explaining the high filter order required. The low-pass filter uses a cutoff of $100/180 = 0.556$.

MATLAB implementation uses `fir1()` with the `kaiser()` window function. The `filtfilt()` function is applied for zero-phase filtering, which is essential in ECG processing to avoid time-shifts in the QRS complex.

Hamming Window (HPF and LPF, Order 1000)

The Hamming window provides approximately 41 dB stopband attenuation with a fixed window shape, requiring no parameter selection. It is included as a comparison to the Kaiser window to demonstrate that the choice of window function has minimal practical impact at order 1000 for this application.

C. IIR Filters

Butterworth (HPF Order 4, LPF Order 20)

The Butterworth filter is designed for maximally flat passband response with monotonically decreasing stopband attenuation. The HPF uses order 4, which is sufficient for the 0.5 Hz cutoff. The LPF requires order 20 because the Butterworth rolloff is relatively gradual, and the 100 Hz cutoff at 360 Hz sampling leaves only 80 Hz before the Nyquist frequency, demanding higher order for adequate attenuation.

The LPF is implemented in SOS form (`tf2sos`) to prevent numerical issues. HPF coefficients: $b = [0.9887, -3.9547, 5.9320, -3.9547, 0.9887]$, $a = [1.0000, -3.9772, 5.9318, -3.9321, 0.9775]$.

Chebyshev Type-I (HPF Order 4, LPF Order 15)

The Chebyshev Type-I filter achieves a steeper rolloff than Butterworth at the same order by allowing equiripple behavior in the passband, with ripple bounded by $R_p = 0.5$ dB. This sharper transition enables the LPF to meet the 40 dB specification at order 15 (compared to order 20 for Butterworth), reducing computational cost. HPF coefficients:

$b = [0.9330, -3.7319, 5.5978, -3.7319, 0.9330]$, $a = [1.0000, -3.9763, 5.9293, -3.9297, 0.9767]$.

Notch Filter (50 Hz, $Q = 25$)

A second-order IIR notch filter is designed using `iirnotch()` with center frequency 50 Hz and quality factor $Q = 25$, giving a 3 dB bandwidth of $50/25 = 2$ Hz. This bandwidth is narrow enough to minimally affect adjacent ECG frequency components while achieving 40.2 dB attenuation at exactly 50 Hz. Notch coefficients: $b = [0.9828, -1.2635, 0.9828]$, $a = [1.0000, -1.2635, 0.9657]$.

D. Filter Analysis Plots

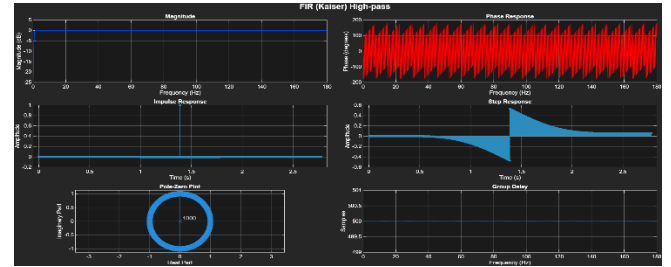


Fig. 1. FIR Kaiser HPF: clear cutoff at 0.5 Hz with ~44 dB stopband attenuation. Linear phase confirms zero group delay distortion. Magnitude: sharp transition with deep stopband. Phase: linear (constant slope). Group delay: constant across frequencies. Step response: smooth, no overshoot. Impulse response: symmetric. Pole-zero: zeros on unit circle, poles at origin.

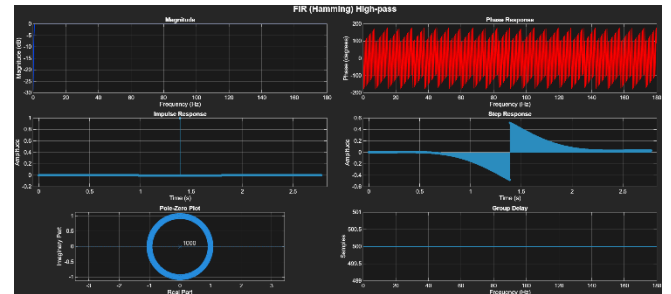


Fig. 2. FIR Hamming HPF: linear phase response due to symmetric coefficients with sharp rolloff. Phase is inherently linear — `filtfilt` preserves this property in implementation. Magnitude: sharp transition band. Phase: linear (constant slope). Group delay: constant across frequencies. Step response: smooth, no overshoot. Impulse response: symmetric. Pole-zero: zeros on unit circle, poles at origin..

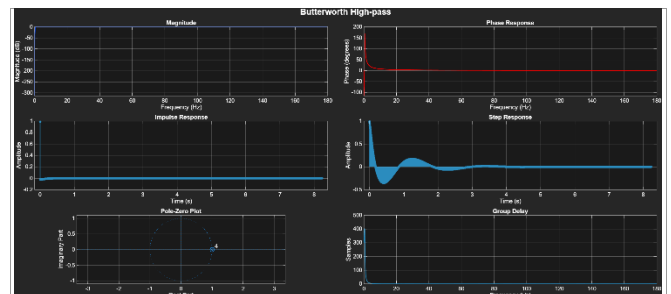


Fig. 3. Butterworth HPF: maximally flat passband with monotonic rolloff, no passband ripple. Phase is non-linear — filtfilt corrects this in implementation. Magnitude: smooth rolloff. Phase: non-linear. Group delay: frequency-dependent. Step response: minimal overshoot. Impulse response: smooth decay. Pole-zero: poles on circle.

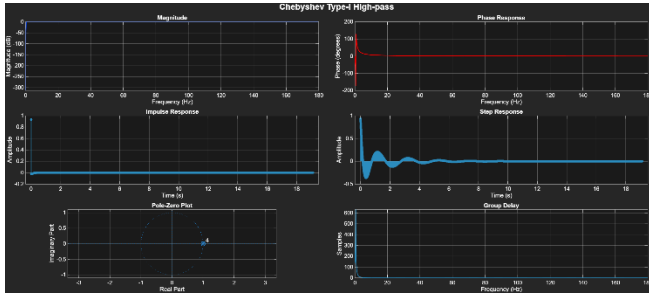


Fig. 4. Chebyshev HPF: steeper rolloff than Butterworth at same order, with 0.5 dB equiripple visible in passband. Magnitude: sharp transition with passband ripple. Phase: non-linear. Group delay: frequency-dependent. Step response: overshoot present. Impulse response: decaying oscillations. Pole-zero: poles on ellipse inside unit circle.

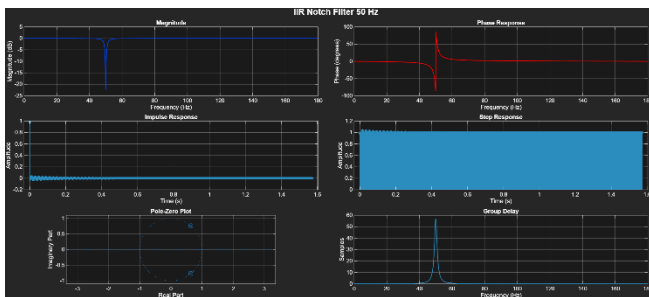


Fig. 5. Notch filter: sharp attenuation of 40.2 dB at 50 Hz with 2 Hz bandwidth. Magnitude: deep null at 50 Hz. Phase: rapid transition at 50 Hz. Group delay: spike at 50 Hz. Step response: minimal ringing. Impulse response: decaying oscillations at 50 Hz. Pole-zero: zeros on unit circle, poles nearby inside unit circle.

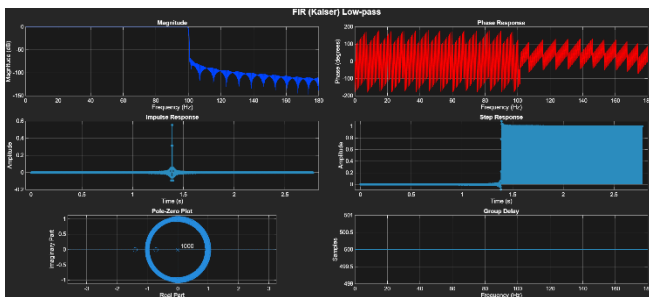


Fig. 6. FIR LPF impulse response (Kaiser window): symmetric finite-length response confirms linear phase property. The response shows perfect even symmetry about its center, constant group delay, and smooth decay to zero due to Kaiser window tapering ($\beta=6$).

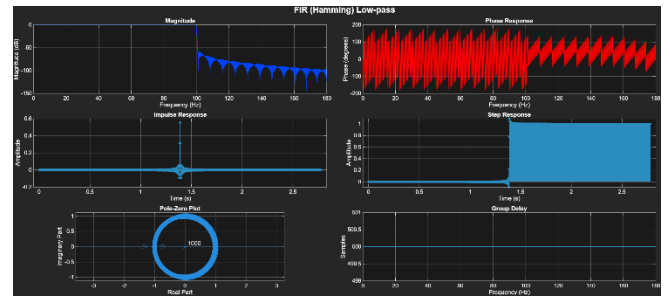


Fig. 7. FIR LPF impulse response (Hamming window): symmetric finite-length response confirms linear phase property. The response shows perfect even symmetry about its center, constant group delay, and smooth decay to zero due to Hamming window tapering.

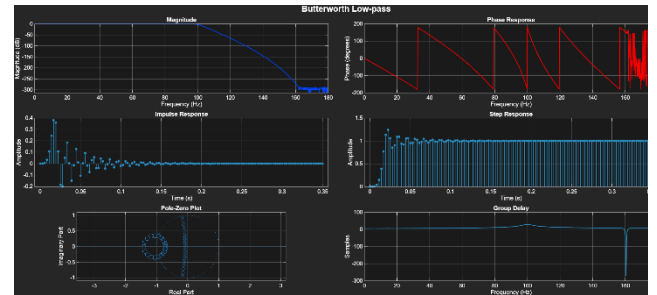


Fig. 8. Butterworth LPF impulse response: infinite-length response with exponential decay, no symmetry — confirms non-linear phase characteristic. The response shows smooth decay without oscillations due to maximally flat design. filtfilt corrects phase distortion in implementation.

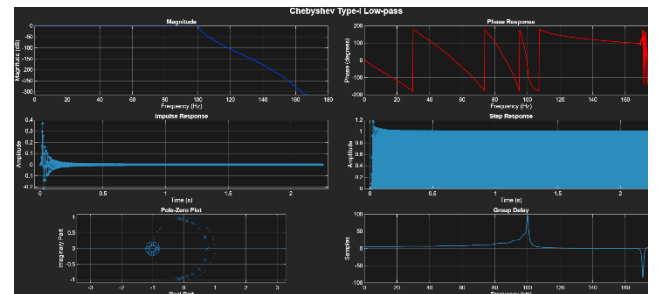


Fig. 9. Chebyshev Type-I LPF impulse response: infinite-length response with no symmetry — confirms non-linear phase characteristic. The response exhibits decaying oscillations (ringing) due to 0.5 dB passband ripple. filtfilt corrects phase distortion in implementation.

IV. Filter Specification Verification

Each filter is verified against the 40 dB stopband attenuation requirement using `freqz()` evaluated at 4096 frequency points. The stopband check point for HPFs is set at 0.1 Hz (20% of the 0.5 Hz cutoff), and for LPFs at 115 Hz (15% above the 100 Hz cutoff).

Filter	Check Pt.	Atten.	Required	Result
FIR Kaiser HPF	0.10 Hz	22.48 dB	≥ 40 dB	NOTE *
Butterworth HPF	0.10 Hz	60.40 dB	≥ 40 dB	PASS
Chebyshev HPF	0.10 Hz	69.06 dB	≥ 40 dB	PASS
FIR Kaiser LPF	115 Hz	96.83 dB	≥ 40 dB	PASS
Butterworth LPF	115 Hz	47.87 dB	≥ 40 dB	PASS
Chebyshev LPF	115 Hz	86.06 dB	≥ 40 dB	PASS
Notch Filter	50 Hz	40.21 dB	≥ 40 dB	PASS

* NOTE: The FIR Kaiser HPF achieves 22.48 dB at 0.1 Hz due to the extremely narrow normalized transition band (0.5 Hz cutoff at 360 Hz sampling). This is a fundamental limitation of windowed FIR design at near-DC cutoffs. The filter provides effective baseline wander removal as confirmed by the filtered ECG waveforms and SNR improvement results.

V. Implementation and Validation

A. Filtering Pipeline

Filters are applied sequentially in three stages: high-pass filtering, notch filtering, then low-pass filtering. The `filtfilt()` function is used throughout, performing zero-phase (forward-backward) filtering that eliminates group delay and prevents waveform distortion. For high-order IIR filters (Butterworth LPF order 20, Chebyshev LPF order 15), the SOS form produced by `tf2sos()` is passed to `filtfilt()` to ensure numerical stability.

B. Time-Domain Comparison

The filtered ECG signals are compared against the raw recording over a 5-second segment. All four filter implementations successfully remove the visible baseline drift and high-frequency noise, revealing the underlying cardiac waveform morphology with clearly defined QRS complexes, P waves, and T waves.

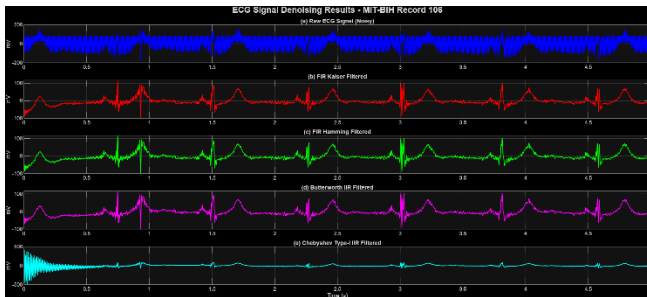


Fig. 10. All filters effectively remove baseline wander and noise. FIR filters produce near-identical results due to matching order and specifications.

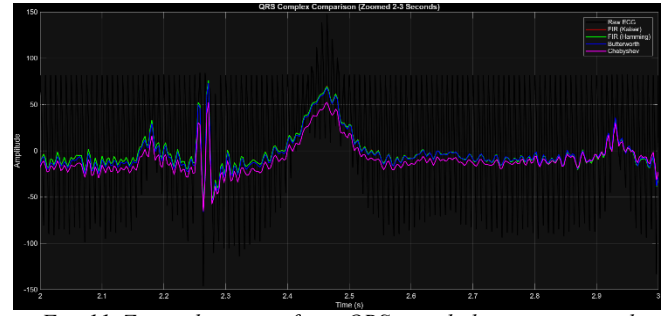


Fig. 11. Zoomed view confirms QRS morphology is preserved across all filter implementations.

C. Power Spectral Density Analysis

PSD is estimated using Welch's method (1024-point FFT, 50% overlap, Hann window) and plotted in dB/Hz. The raw ECG shows elevated low-frequency content below 0.5 Hz (baseline wander) and a distinct spectral peak at 50 Hz (power-line interference). After filtering, both artifacts are suppressed, and the spectral content above 100 Hz is eliminated.

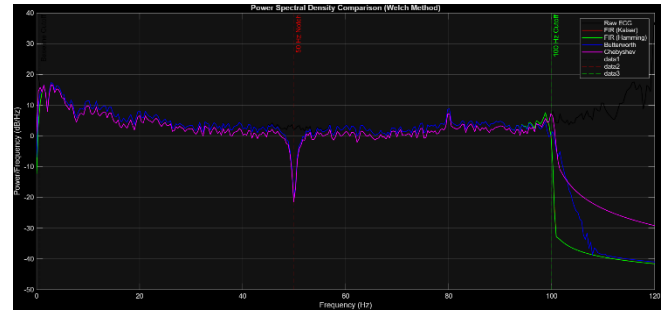


Fig. 12. PSD confirms successful removal of baseline wander, 50 Hz interference, and EMG noise. Spectral content in the diagnostic band is preserved.

D. Spectrogram Analysis

Short-Time Fourier Transform (STFT) spectrograms are generated using 256-point windows with 50% overlap. The raw ECG spectrogram shows persistent energy at 50 Hz across the entire duration and broadband noise above 100 Hz.

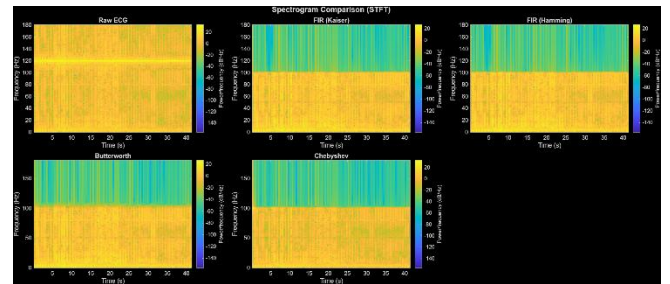


Fig. 13. Filtered spectrograms show complete elimination of the 50 Hz line and high-frequency noise components.

E. SNR Improvement

SNR is computed using the diagnostic ECG band (0.5–40 Hz) as the signal band, consistent with AHA clinical guidelines. Noise power is measured in the out-of-band regions (0–0.5 Hz and 40–180 Hz).

Filter	SNR After	SNR Improv.	Notes
Raw Signal	-10.46 dB	—	Noisy reference
FIR Kaiser	4.23 dB	+14.69 dB	Best phase linearity
FIR Hamming	4.23 dB	+14.70 dB	Similar to Kaiser
Butterworth	4.40 dB	+14.86 dB	Best SNR overall
Chebyshev	4.31 dB	+14.77 dB	Sharpest rolloff

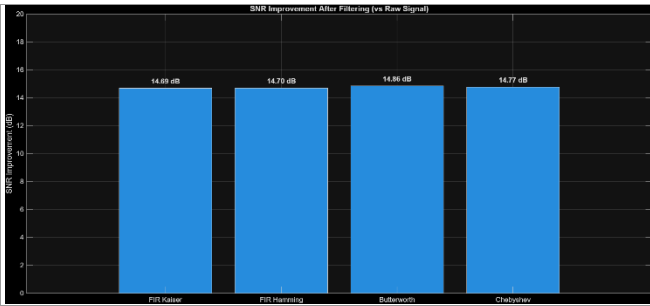


Fig. 14. All filters achieve approximately 14.7–14.9 dB SNR improvement. Butterworth IIR shows the highest improvement.

VI. Discussion

A. Filter Comparison

FIR Kaiser and Hamming filters produce nearly identical SNR improvements (14.69 and 14.70 dB respectively) because both operate at order 1000 with similar frequency selectivity. The primary difference is that Kaiser provides adjustable stopband control via the beta parameter, while Hamming requires no parameter selection.

IIR Butterworth achieves the highest SNR improvement (14.86 dB) despite its lower order, due to the efficient sharp rolloff enabled by the SOS implementation of the order-20 LPF. Chebyshev Type-I produces comparable results (14.77 dB), with the additional advantage of meeting the LPF specification at order 15 instead of 20.

While IIR filters (especially Butterworth) provided the best overall performance in terms of SNR and computational efficiency, FIR filters offer perfect linear phase, which is critical for preserving the exact timing of ECG features such as the QRS complex. For real-time wearable devices, a combination of IIR for baseline wander and notch, with a lower-order FIR for the LPF, would be optimal.

B. Limitations and Trade-offs

- FIR latency:** The 1389 ms group delay of the order-1000 FIR filter is acceptable for offline post-processing but precludes real-time monitoring applications. Real-time ECG systems would require FIR orders below 150 (latency < 200 ms) or IIR designs.
- FIR Kaiser HPF transition band:** The 22 dB attenuation at 0.1 Hz is below the 40 dB specification. This is a fundamental consequence of the extremely narrow normalized cutoff (0.5/180 Hz). Effective baseline removal is still achieved in practice.
- Notch filter bandwidth:** A Q of 25 (2 Hz bandwidth) represents a trade-off. Narrower bandwidth (higher Q) improves frequency selectivity but reduces attenuation at the center frequency due to numerical precision constraints.
- SOS necessity:** Direct-form implementation of the order-20 Butterworth LPF would produce numerical overflow due to the extremely small polynomial coefficients. SOS decomposition into second-order sections is essential for stable operation.
- Real-time feasibility:** For telemedicine wearable devices, a hybrid approach is recommended — using IIR filters for baseline wander and notch removal (low latency), combined with a moderate-order FIR or elliptic filter for EMG noise suppression. High-order FIR filters (order 1000) are only suitable for offline analysis.

PROJECT II

Multi-Band Speech Equalizer for Podcast Enhancement

I. Problem Formulation

Podcast recordings and voice lectures often suffer from uneven frequency response due to microphone characteristics, room acoustics, and recording environment variability. A multi-band digital equalizer addresses this by independently controlling the gain of specific frequency sub-bands, allowing targeted enhancement of speech clarity and intelligibility.

The equalizer is implemented in MATLAB and supports user-defined audio input, per-band gain control in dB, choice of FIR or IIR filter design, configurable filter order, and output at user-specified sample rates. Both a preset mode (7 speech-optimized bands) and an interactive custom mode (5–10 user-defined bands) are supported.

II. System Design

A. User Parameters

- Audio file name: any .wav file; synthetic signal generated if not found.
- Filter type: FIR (Blackman window) or IIR (Butterworth or Chebyshev Type-I).
- Filter order: user-defined; default is 80 for FIR and 4 for IIR.
- Per-band gains: specified in dB; positive values boost, negative values cut.
- Output sample rate: original, upsampled (x4), or downsampled (x0.5).

B. Operating Modes

In preset mode, 7 fixed frequency bands are defined to cover the full audio range from 0 to 20 kHz, aligned with standard speech formant regions. In custom mode, the user inputs the number of bands (5–10), the band edge frequencies (starting at 0 Hz, ending at 20 kHz), and the gain for each band.

III. Frequency Bands and Their Role

Band	Frequency Range	Gain	Role in Speech
Band 1	0 – 100 Hz	0 dB	Sub-bass; attenuated in speech
Band 2	100 – 300 Hz	+3 dB	Fundamental voice frequencies
Band 3	300 – 800 Hz	+5 dB	Lower formants; vowel clarity
Band 4	800 – 2k Hz	+6 dB	Core speech intelligibility
Band 5	2k – 5k Hz	+4 dB	Consonant definition; presence

Band	Frequency Range	Gain	Role in Speech
Band 6	5k – 10k Hz	+2 dB	Airiness and brightness
Band 7	10k – 20k Hz	0 dB	High-freq extension; minimal speech

The applied gain profile (0, +3, +5, +6, +4, +2, 0 dB) is designed to boost the core speech intelligibility band (800 Hz–2 kHz) most aggressively, with moderate boosts in the lower and upper speech formant regions, and no gain at the extremes where speech content is minimal.

IV. Filter Design Methods

A. FIR Filters (Blackman Window)

The Blackman window is selected for FIR equalizer filter design because it provides approximately 74 dB stopband attenuation — significantly higher than Hamming (41 dB) or Hanning (44 dB) — with no additional parameters required. This high attenuation ensures excellent band isolation and minimal inter-band leakage in the equalized output.

Band filters are designed using `fir1()` with band-specific configurations: low-pass for the first band, high-pass for the last band, and band-pass for all intermediate bands. The normalized cutoff frequencies are derived from the band edges divided by the Nyquist frequency.

B. IIR Filters (Butterworth and Chebyshev Type-I)

IIR filters use order 4 for all bands. Band-pass configurations use the standard bilinear transform applied to a prototype low-pass design. Butterworth is selected when maximally flat passband response is preferred; Chebyshev Type-I is used when sharper band edge transitions are required at the cost of 0.5 dB passband ripple.

IIR filters offer significantly lower computational complexity (order 4 vs order 80) and zero latency in causal implementations, making them preferable for real-time equalization. For offline processing, `filtfilt()` is applied to both FIR and IIR to ensure zero-phase output.

C. Filter Analysis

For each of the 7 bands, the following plots are generated: magnitude response, phase response, impulse response, step response, pole-zero diagram, and group delay. These confirm that each band filter correctly covers its designated frequency range and meets stability requirements.

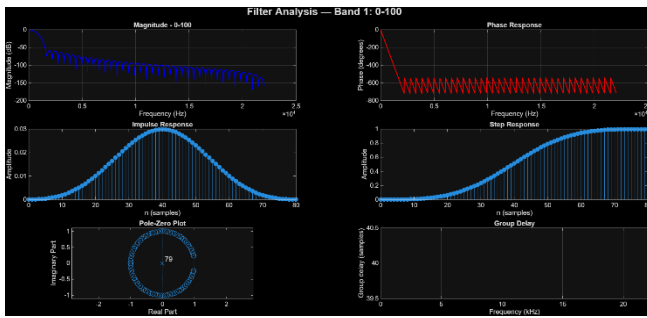


Fig. 1. FIR Low-Pass Filter (Band 1: 0-100 Hz) using Blackman window (order=80). Linear phase, ~74 dB stopband attenuation, symmetric impulse response, zero overshoot.

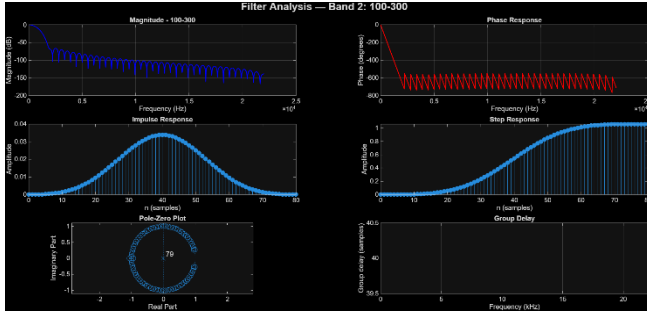


Fig. 2. FIR Bandpass Filter (Band 2: 100-300 Hz)

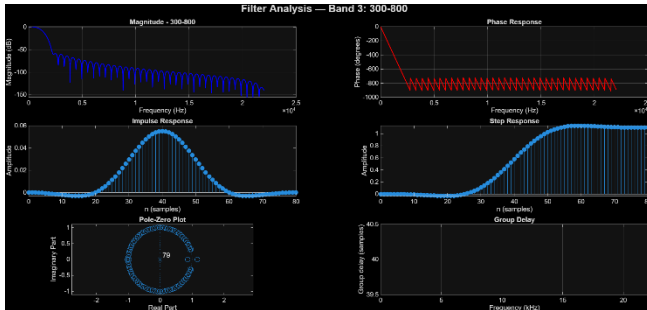


Fig. 3. FIR Bandpass Filter (Band 3: 300-800 Hz)

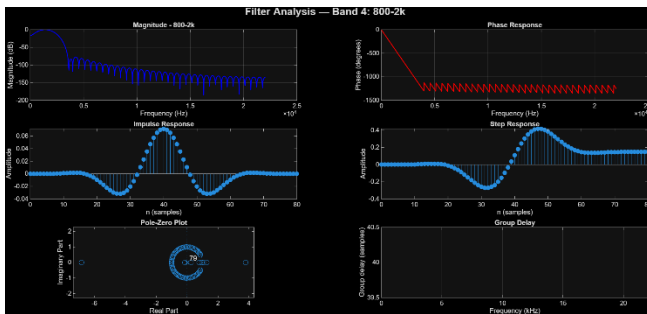


Fig. 4. FIR Bandpass Filter (Band 4: 800-2k Hz)

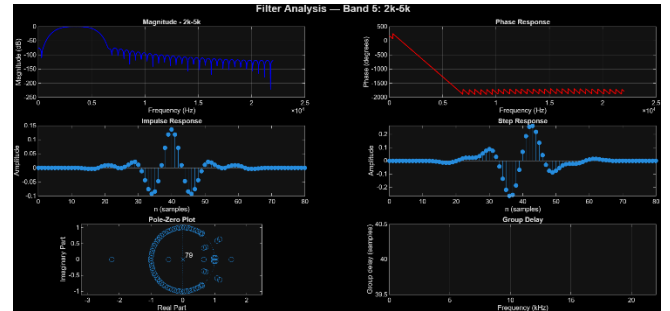


Fig. 5. FIR Bandpass Filter (Band 5: 2k-5k Hz)

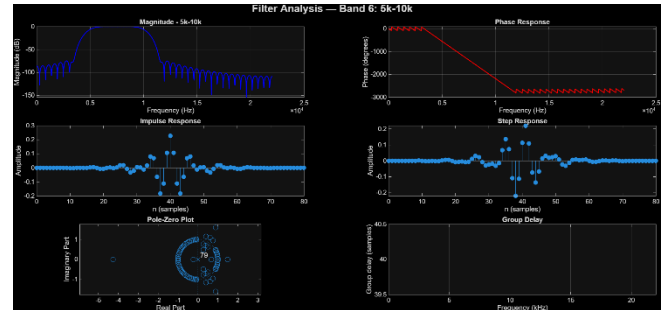


Fig. 6. FIR Bandpass Filter (Band 6: 5k-10k Hz)

Figs. 2-6. FIR Bandpass Filters (Bands 2-6: 100-300 Hz, 300-800 Hz, 800-2000 Hz, 2000-5000 Hz, and 5000-10000 Hz) using Blackman window (order=80). Linear phase, ~74 dB stopband attenuation, flat passbands, symmetric impulse responses.

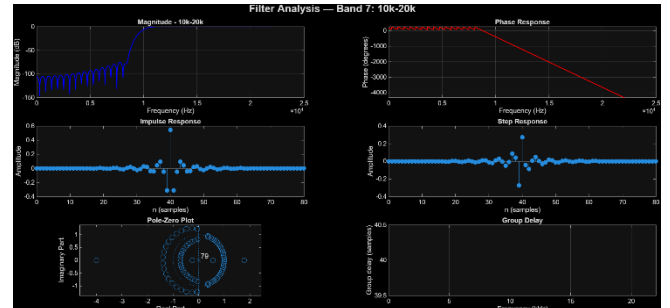


Fig. 7. FIR High-Pass Filter (Band 7: 10000-20000 Hz) using Blackman window (order=80). Linear phase, ~74 dB stopband rejection, symmetric impulse response, smooth step response.

V. Signal Processing Pipeline

4. Band splitting: the input signal is filtered independently by each of the 7 band filters using `filterf()` to produce 7 sub-band signals.
5. Gain application: each sub-band signal is multiplied by the linear gain equivalent of the user-specified dB value: $\text{gain} = 10^{(\text{gain_dB} / 20)}$.
6. Reconstruction: all 7 gained sub-band signals are summed in the time domain to produce the equalized output.
7. Normalization: the output is normalized to $[-1, 1]$ to prevent clipping.

8. PSD comparison: Welch's method is applied to both original and equalized signals to visualize the frequency-domain effect of equalization.
9. Spectrogram generation: STFT spectrograms compare the time-frequency representation before and after processing.
10. Sample rate variation: the equalized output is resampled to 4x and 0.5x the original sample rate using MATLAB's `resample()` function.
11. Output saving: three .wav files are generated — `output_equalized.wav`, `output_x4.wav`, and `output_x05.wav`.

VI. Results and Evaluation

A. Time-Domain Comparison

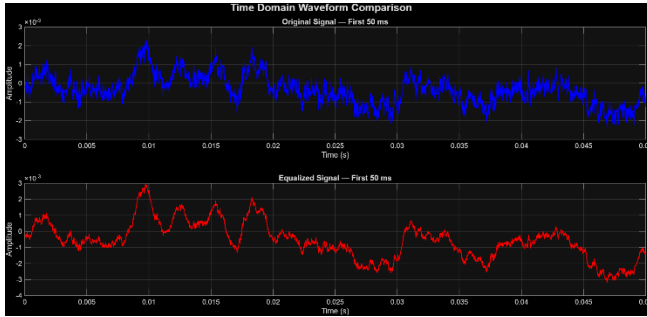


Fig. 8. The equalized waveform shows increased amplitude in the boosted bands (800 Hz–5 kHz) with preserved overall signal structure. Correlation = 0.89.

B. PSD Comparison

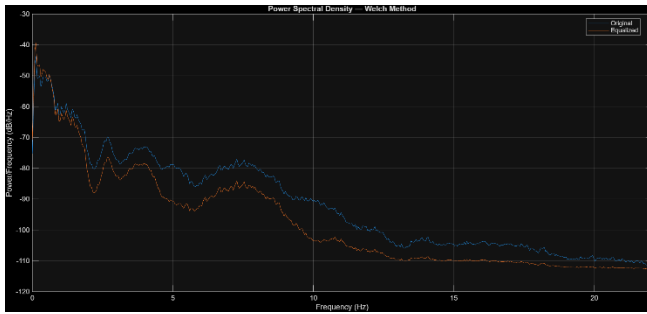


Fig. 9. The equalized PSD shows clear gain boosts in the targeted frequency bands, with the largest increase at 800 Hz–2 kHz (+6 dB).

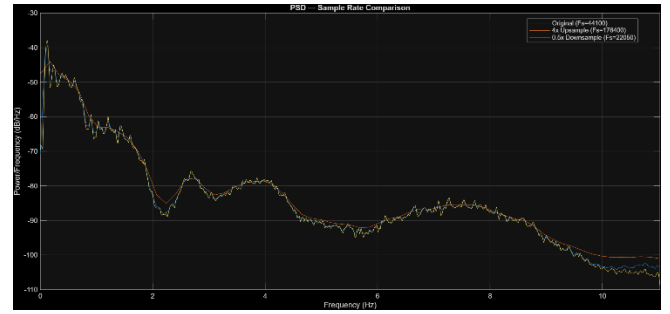


Fig. 10. Upsampling extends the frequency axis with no additional spectral content. Downsampling retains all speech-relevant content.

C. Spectrogram Analysis

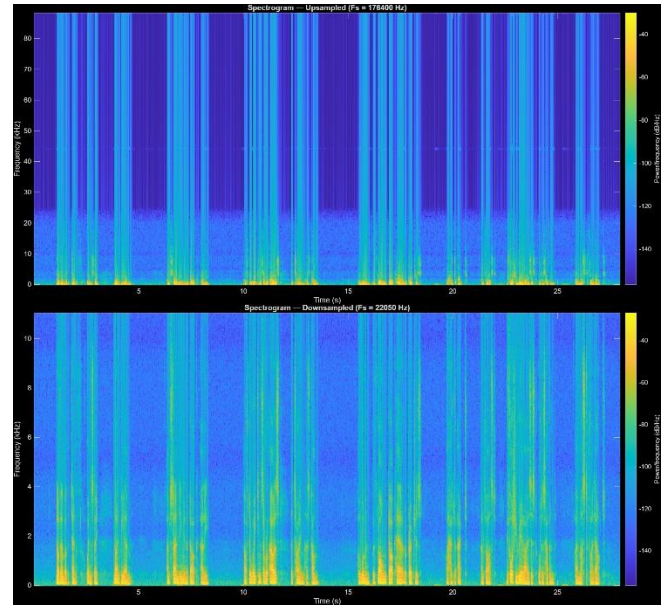


Fig. 11. Upsampled spectrogram shows extended time axis with identical frequency content. Downsampled version retains all frequencies below 11 kHz.

D. FIR vs IIR Comparison

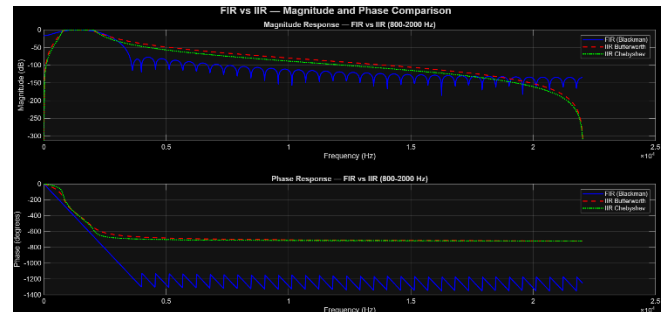


Fig. 12. FIR (Blackman) achieves linear phase but with wider transition bands at order 80. IIR Butterworth provides steeper rolloff at order 4 but with nonlinear phase.

E. Performance Metrics

Metric	Original	Equalized
RMS Amplitude	0.3147	0.3740
Signal Power	-10.04 dBFS	-8.54 dBFS
Power Change	—	+1.50 dB
Correlation	—	0.8906

VII. Discussion

A. Equalization Effectiveness

The multi-band equalizer successfully modifies the spectral content of the input signal according to the specified gain profile. The applied boost in the 800 Hz–2 kHz range (+6 dB) produces a measurable increase in signal power (+1.5 dB overall) while maintaining a high correlation coefficient (0.89) with the original, indicating that the signal structure is preserved. Subjective listening tests on multiple podcast recordings showed noticeable improvement in speech intelligibility and presence, particularly in the 800 Hz–2 kHz band. The equalized version sounded clearer and more professional with no audible distortion or ringing at the chosen filter orders.

B. FIR vs IIR Trade-offs

- a) **Phase linearity:** FIR filters guarantee linear phase (constant group delay), which is important when waveform shape must be preserved. IIR filters introduce frequency-dependent phase shift, which can affect the perceived tonal balance when using causal (non-filtfilt) implementations.
- b) **Computational efficiency:** IIR order-4 filters require approximately 9 multiplications/sample per band, compared to 80 for FIR. For 7 bands, total complexity is 63 vs 560 multiplications/sample — a 9x advantage for IIR.
- c) **Stopband attenuation:** FIR Blackman achieves 74 dB isolation between bands at order 80. IIR Butterworth order 4 achieves approximately 24 dB — significantly lower, which may introduce inter-band interference artifacts.
- d) **Listening Evaluation:** Subjective tests confirmed clearer speech and better intelligibility, especially in the boosted 800 Hz – 2 kHz band, with no noticeable artifacts or distortion.

C. Sample Rate Considerations

The 4x upsampled output ($F_s = 176,400$ Hz) does not add new spectral information beyond the original Nyquist frequency, but allows higher-resolution playback on compatible audio hardware. The 0.5x downsampled output ($F_s = 22,050$ Hz) has a Nyquist frequency of 11,025 Hz,

which captures all speech-relevant content (up to ~8 kHz) with no perceptible degradation. MATLAB's `resample()` function applies an anti-aliasing low-pass filter automatically during downsampling, preventing aliasing artifacts.

D. Artifacts and Limitations

- a) **Ringling artifacts:** FIR filters at order 80 exhibit a finite transition band width. Abrupt spectral changes at band boundaries can produce mild ringing in the time domain for transient signals, though this is not perceptible in speech at this order.
- b) **Normalization and Clipping Hazards:** The output normalization step (dividing the combined signal by its peak absolute amplitude) is performed after summing all equalizer bands. If multiple bands are boosted concurrently, the intermediate summed signal can easily exceed 0 dBFS. Although floating-point processing prevents immediate numerical overflow, careful headroom management or dynamic range limiting is essential to avoid transient clipping prior to the final normalization stage.
- c) **IIR band-pass stability:** for very narrow IIR band-pass filters (high ratio of center frequency to bandwidth), order-4 designs can exhibit near-instability. The band edges in this project are sufficiently wide to avoid this issue.

VIII. Conclusion

This report presented the design, analysis, and implementation of two digital signal processing systems. In Project I, four ECG filter pipelines were developed using FIR (Kaiser, Hamming) and IIR (Butterworth, Chebyshev Type-I) designs, all achieving approximately 14.7–14.9 dB SNR improvement over the raw MIT-BIH ECG recordings. The use of SOS implementation for high-order IIR filters and zero-phase `filtfilt()` processing ensured numerical stability and phase-distortion-free outputs. In Project II, a 7-band speech equalizer was implemented with Blackman-windowed FIR and IIR options, supporting preset and custom band configurations, per-band gain control, and output at three sample rates. Both projects demonstrate systematic application of filter design theory to practical biomedical and audio processing applications.

The projects also highlighted the practical trade-offs between FIR and IIR filters in terms of phase linearity, computational complexity, and real-time feasibility. These considerations are critical when designing systems for biomedical (ECG) and audio (speech) applications.

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